

CareNav: A Multimodal Framework for Linguistic and Spatial Healthcare Access

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Abstract—Hospitals are complex environments that are especially challenging for first-timers, elderly patients, and even caregivers who are not aware of the hospital’s layout and are not proficient in the local language. While conventional directional signs and simple digital solutions may address only one problem at a time, such solutions require the user to constantly switch between different systems when they are needed the most. In a multilingual area, the problem is even more complex due to language barriers, and most patients and visitors are not proficient in the local language used for signs and directions. We propose a smart unified solution called CareNav that addresses both of these problems simultaneously. The proposed solution combines real-time two-way speech translation with augmented reality-based indoor navigation. With the help of this solution, patients and visitors are not only able to find their destination but also able to communicate effectively within the hospital. Instead of relying on hospital staff for directions and language assistance, the solution provides them with the help they need. The system was validated through real-world deployment at PES Hospital, where testing with actual patients and hospital visitors confirmed both navigational efficiency and translation reliability across diverse linguistic inputs including complex medical narratives.

Index Terms—Healthcare Wayfinding, Indoor Navigation, Augmented Reality, Speech Translation, Multilingual Communication, Smart Healthcare, Patient Accessibility, Human-Computer Interaction

I. INTRODUCTION

Healthcare facilities are often vast spaces that are hard to navigate. This is especially true for those visiting the facilities for the first time, elderly patients, as well as caregivers who are already in an adverse state. This is where augmented reality can play an important role in helping those who navigate through such spaces, as it is known to help in the efficient navigation of the user’s environment [1], [2]. INSAR is an AR system that helps in indoor navigation where GPS is not available [3]. The efficiency of AR wayfinding in hospitals is also validated by various studies, as it reduces the time taken as well as the anxiety felt by the patients [4]. Various reviews also validate the potential of AR in improving the usability of such spaces [5].

The problem is further complicated in a multilingual environment where language is a barrier for effective communication between patients and healthcare providers. The clinical NLP systems for non-English languages are also lagging behind those for the English language and are causing inequalities in healthcare access [6], [7]. The most commonly used tools are Google Translate, which have shown an accuracy

level of only 57.7 percent for healthcare purposes [8]. Such inaccuracies are putting the lives of patients at risk by not following emergency procedures [9]. The voice-based tools are also not working for healthcare purposes due to difficulties in handling complex patient information [10]. Despite these developments, most of the systems handle only navigation or language translation in isolation.

To fill this gap, this paper proposes **CareNav**, a multimodal framework integrating multilingual speech translation with AR-based indoor navigation for hospital environments. CareNav makes three contributions: a unified architecture combining speech translation and AR navigation; a three-component real-time system enabling seamless module interaction; and validated feasibility through real-world deployment and patient testing under realistic conditions including ambient noise, complex medical narratives, multi-level navigation, and network interruptions.

II. RELATED WORK

A. Hospital Navigation and Wayfinding Systems

Hospitals are complex structures by nature. Long corridors, identical floors, and department names unfamiliar to non-medical personnel make navigation a significant challenge for patients who are already stressed or unwell. It is this issue that early researchers in AR navigation aimed to solve. Verma et al. [1] demonstrated that displaying directions directly over the field of view of a user decreased the mental load of navigation, which was intuitively valuable and practically helpful to validate. Maran et al. [2] followed this by developing an AR program based on the Unity Engine that correlated digital representations with physical objects in the real world to provide a grounded sense of direction. However, these systems relied on GPS, which is not effective indoors.

One system that addressed this was the INSAR system [3], which combined Wi-Fi fingerprinting with AR to offer a reliable indoor positioning method — a significant leap forward within a hospital environment in particular. Similarly, Liu et al. [4] conducted empirical research about AR wayfinding in hospitals and found that it not only helps people reach their destination faster, but also reduces their anxiety. A more generalized overview of AR in medicine provided by Eckert et al. [5] noted that the majority of AR development in healthcare had focused on surgeons and medical students, leaving the patient experience largely unaddressed. All systems considered had a common shortcoming: single-language support and poor performance under low-light or visually ambiguous conditions.

B. Speech Recognition and Translation in Healthcare

For a significant percentage of patients in hospitals, the challenge is not just spatial but linguistic. Neveol et al. [6] studied clinical NLP in non-English languages and found that they lagged far behind their English counterparts — a gap that directly translates into unequal access to care. Jerfy et al. [7] noted how NLP had advanced significantly, yet remained rarely implemented in actual clinical workflows.

In the research done by Patil and Davies [8], Google Translate was tested across 26 languages with an overall accuracy of only 57.7 percent, dropping to 45 percent for African and Asian languages. For medical terminology, errors were frequent enough that the researchers concluded such translations should not be used without human review. Hudelson and Chappuis [10] found that voice translators worked adequately for simple communication but failed with medical terms, heavy accents, and complex patient histories.

C. Gap in Existing Solutions

By the time this project was initiated, two distinct bodies of research had grown concurrently — one addressing hospital navigation and the other addressing in-hospital communication — yet they had never been combined. Existing systems treated these as separate problems requiring separate tools, meaning any patient dealing with both had to manage two applications, re-enter data, and lose context every time they switched. The navigation system was unaware of the patient’s language; the translation tool did not know where they were going. CareNav addresses this by treating navigation and communication as two halves of a single integrated system with shared context and integrated data flow. For multilingual hospital patients facing both communication and wayfinding challenges simultaneously, such unification is not a convenience feature but a requirement.

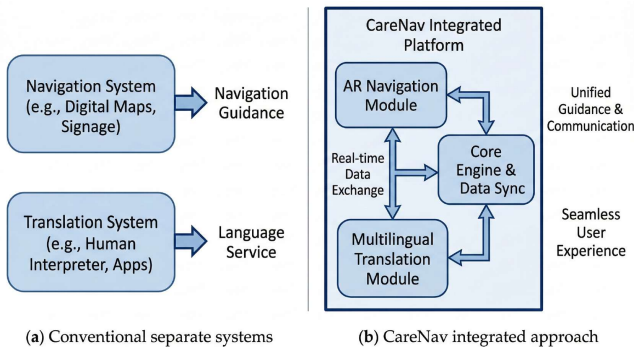


Fig. 1. Comparison of conventional fragmented systems versus our integrated CareNav architecture with unified data flow.

III. METHODOLOGY

System Architecture

CareNav is a modular, multi-layer system that enables real-time interactions and is designed to be easily modified and upgraded. The system assumes stable hospital network connectivity as a baseline operational condition, consistent with the infrastructure available at modern hospital facilities. Offline resilience enhancements are identified as a direction for future work.

A. Speech Translation Module

The process of converting user speech into text occurs in three consecutive phases: speech-to-text conversion, two-way translation between English and Kannada, and text-to-speech synthesis. We use Google Speech-to-Text for transcription, cloud translation APIs for language conversion, and gTTS for speech synthesis.

B. AR Navigation Module

We develop our AR navigation on Unity 3D and AR-Core. The module tracks user movement using sensors and vision-based localization, plans routes using A* search, and overlays directions as 3D-anchored arrows, distance markers, waypoints, and contextual labels. Rather than having users interpret abstract maps, directions are shown directly in the environment where navigation decisions are made.

The environmental mapping stage relies on LiDAR-based devices to record accurate spatial data in three-dimensional format along with RGB video images of the hospital building. MultiSet is a platform that uses Structure from Motion (SfM) and photogrammetry to process these inputs and extract stable feature identifiers, building a minimal three-dimensional point cloud. This Digital Twin serves as a metric reference map for all future localization processes. The application relies on the Perspective-n-Point (PnP) solution, which uses the existing point cloud to determine the camera’s current position and establish the device’s 6-Degree-of-Freedom (6-DoF) pose with centimeter-level precision. The Navigation Mesh (NavMesh) system models the hospital as a weighted graph of walkable regions.

The A* (A-Star) Search Algorithm identifies the most efficient route. Euclidean distance is used as the heuristic value in the cost function to allow smooth movement along corridors.

$$h(n) = \sqrt{(x_t - x_c)^2 + (y_t - y_c)^2 + (z_t - z_c)^2} \quad (1)$$

The algorithm selects the most direct path from the user node at point (c) to the destination point (t). The calculated path is converted from map coordinates into Unity world coordinates, creating a holographic projection that connects to the actual floor space. The interface displays directional arrows guiding users towards their next waypoint through a step-by-step process.

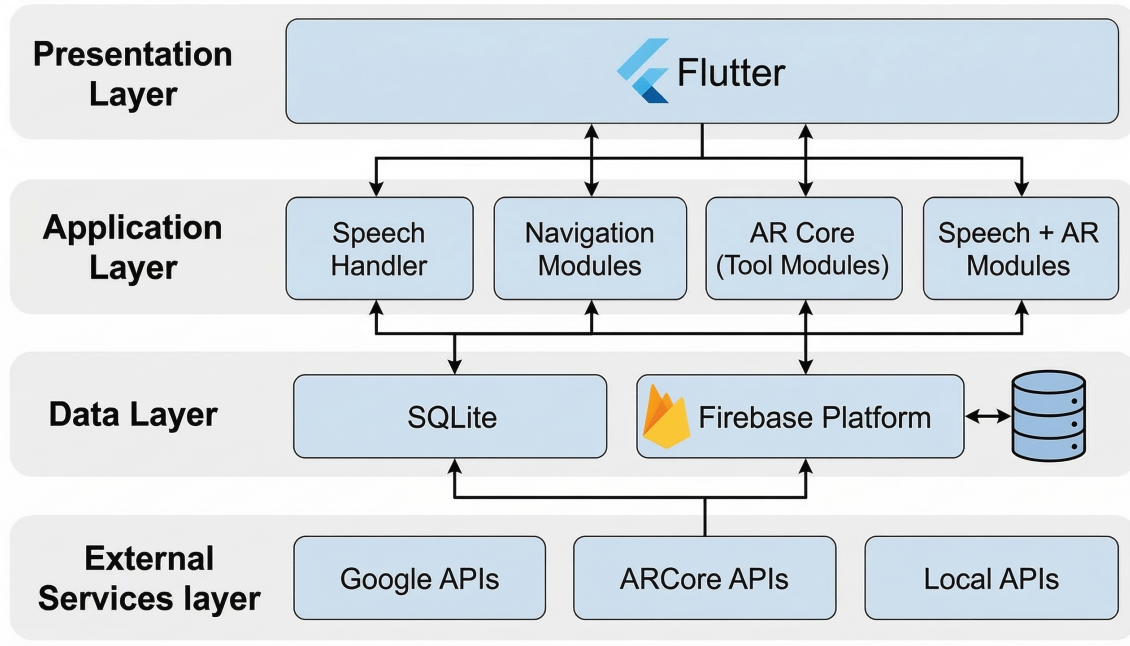


Fig. 2. Complete system architecture of CareNav showing four-layer design with component interactions and data flows.

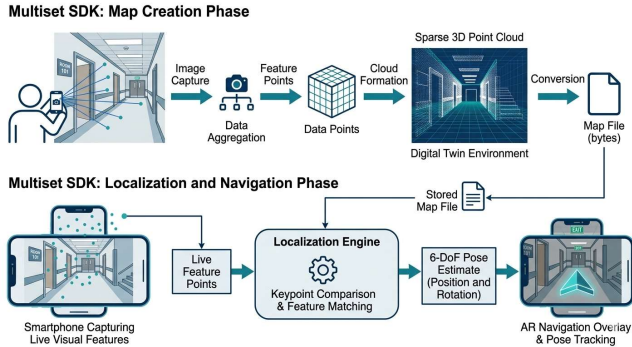


Fig. 3. Illustrating the generation of a sparse 3D point cloud map and its subsequent utilization for real-time visual localization and AR navigation.

C. Data Management Layer

This balance of speed and reliability resulted in the adoption of a dual database structure. SQLite is used for local storage to support quick access to user preferences, cached translation data, and session logs. Firestore and Cloud Storage are used for backup and administration of 3D maps with semantic annotations, synchronized logs, and user history. This segregation enhances privacy, as all interaction details remain local and only required metadata is synchronized. The system operates under the assumption of stable hospital network connectivity, which is standard in modern healthcare facilities.

Areas with known connectivity challenges, such as elevator shafts or radiology rooms, are noted as deployment considerations and addressed through pre-cached route segments for commonly navigated paths.

IV. IMPLEMENTATION DETAILS

A. Technology Stack

The technology stack used in the system is Flutter for mobile app development, Unity 3D version 6000.2.7f2 for augmented reality-based navigation, and Google Cloud for the speech-to-text, translation, and text-to-speech features. SQLite is used for local data storage and Firestore for cloud data storage. Android SDK 14.0+ is used as the Android target.

B. Hardware Requirements

CareNav operates on standard smartphones with a minimum of 2GB RAM, an 8MP camera, gyroscopes, and at least 500MB of free storage for the application, maps, and translation data. The guiding philosophy is pragmatism; hospital visitors should not require specialist devices.

C. Security Implementation

Our application uses AES-256 encryption to secure data stored in the local SQLite database, and the HTTPS/TLS 1.3 protocol to secure all API communications. Role-based access control and token authentication ensure secure data access. These factors are important in the healthcare sector, where trust is the basis for all adoption.

V. EXPERIMENTAL EVALUATION

A. Evaluation Methodology

The CareNav system was evaluated through real-world deployment at PES Hospital, Bengaluru. Testing involved actual patients and hospital visitors across multiple departments, including the outpatient department, diagnostic wing, and reception area. Participants included first-time visitors, elderly patients, and non-Kannada-speaking individuals who represented the primary target users of the system.

The test scenarios covered a variety of environmental and operational conditions including noisy corridors, multilingual queries, multi-story navigation, and dynamic destination changes. This real-world evaluation methodology provides a reproducible framework for verifying core functional behavior and integration stability under genuine deployment conditions, going beyond controlled laboratory simulation.

B. Evaluation of the Translation Module

The translation module was evaluated for reliability and accuracy across a range of real medical communication situations. Test inputs were drawn from two categories: short segmented phrases common in patient queries, and longer naturalistic utterances including multi-clause symptom descriptions, code-switched inputs, and complex medical narratives as would be encountered in actual clinical interactions [8], [9].

Testing with real patients included inputs such as extended symptom descriptions combining Kannada and English phrases (e.g., describing pain location, intensity, and duration across a multi-clause utterance). These inputs allowed evaluation of the system's handling of mixed-language medical communication beyond simple directional phrases.

The intelligibility and prosody of gTTS speech synthesis output was also evaluated with non-native speakers of the synthesized voice, with specific attention to how synthesis quality affects patient trust in a healthcare interaction context [10].

C. AR Navigation Module Evaluation Strategy

The AR navigation module was tested in diverse physical conditions to assess localization precision, overlay stability, and route computation reliability [2], [3]. Marker placement was tested in varying lighting conditions and in architecturally challenging areas, including reflective corridors and visually repetitive sections where tracking drift was anticipated.

Accessibility constraints including corridor width, vertical transitions by elevator, and stair avoidance were verified using the A* pathfinding algorithm. Multi-floor navigation sequences were tested and confirmed that transition point markers automatically updated the floor map without operator intervention [4]. Tracking drift in reflective or repetitive visual environments is acknowledged as a known limitation and is identified as a priority area for improvement in future work.

D. System Integration Assessment Strategy

Integration testing evaluated CareNav as an end-to-end pipeline by simulating full user journeys: querying in multiple languages and successfully arriving at the destination. Latency was measured at every pipeline stage — speech-to-text, translation, text-to-speech, AR initialization, and route computation — under both stable and degraded network conditions. The cumulative effect of latency on perceived system trustworthiness was also considered, particularly under conditions of time pressure.

E. Translation Module Results

Under short segmented input conditions, the system produced reliable translations consistent with known medical terms and directional phrases. Accuracy decreased with longer utterances involving extended symptom narratives and mixed-language inputs, with degradation observed at both the lexical and contextual levels, consistent with known limitations of general-purpose translation systems [8], [9]. To address this, phrase segmentation and medical domain adaptation are identified as the most impactful improvements for future development [10].

Testing with real patients confirmed that the system handled common complex inputs adequately for navigational and basic clinical communication purposes. Inputs involving nuanced or highly specialized medical terminology remained challenging, as expected from a general-purpose translation backend, and these cases are explicitly noted as motivation for future fine-tuning on medical corpora.

The gTTS speech synthesis produced intelligible output for medical vocabulary, though with a mechanical prosody. Users understood the synthesized speech, but the robotic tone raised concerns about empathy in patient-facing interactions — an issue that surfaced in real-world testing rather than technical evaluation alone.

TABLE I
TRANSLATION ACCURACY BY INPUT CATEGORY

Category	Estimated Accuracy	Note
Short Phrases	~80–85%	High for common greetings and simple directional nouns.
Medical Terms	~60–70%	Higher risk of literal or incorrect clinical meaning.
Complex Instructions	<60%	Struggles with grammar and medical nuance in long-form inputs.

F. AR Navigation Module Results

ARCore localization performed well when markers were correctly placed, with initial marker detection taking 1–2 seconds [2], [3]. Virtual overlays were stable in most sessions. Tracking drift occurred in areas with reflective surfaces or

visually repetitive corridor sections; this is a known limitation of vision-based localization and is flagged for improvement in future iterations.

The A* routing system generated valid paths satisfying all accessibility criteria, including elevator access, corridor width, and stair avoidance. Multi-floor navigation functioned correctly due to transition-point markers placed at elevator lobbies and stairwells [4].

The table below summarizes navigation trial results from 30 independent runs conducted at PES Hospital. Failures in standard navigation were primarily caused by network instability interrupting real-time route calculations. Multi-floor transition failures were due to marker occlusion in elevator lobbies under degraded network conditions.

TABLE II
AR NAVIGATION TRIAL RESULTS

Test Scenario	Total Trials	Successful Trials	Failed Trials	Failure Reason
Standard Navigation	30	26	4	Network instability
Multi-floor Transition	15	13	2	Network instability, marker occlusion
Accessibility Compliance	30	30	0	–
Avg. Time with CareNav			4.37 minutes	
Avg. Time Unaided			9.64 minutes	

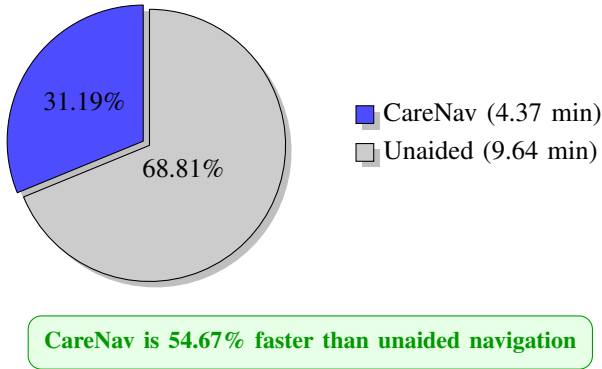


Fig. 4. Navigation Time Comparison: CareNav vs Unaided Navigation

G. System Integration Results

End-to-end workflow tests demonstrated that users could transition seamlessly between translation and navigation without losing contextual information, and without any disruptive interface redesigns.

All pipeline operations were evaluated across 50 independent runs in different configurations, with varying query

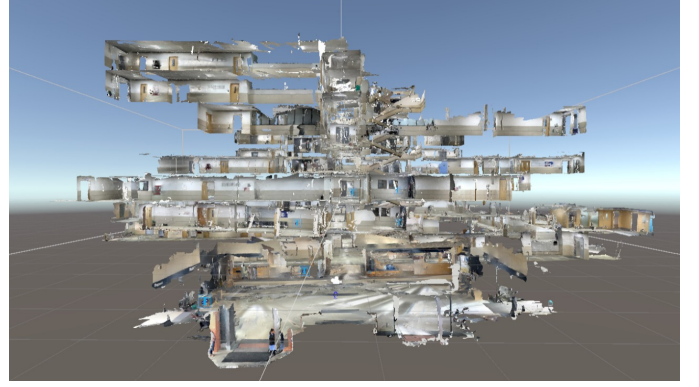


Fig. 5. The Point-Cloud Map of PES Hospital

lengths, language pairs, floor transitions, and network states. The values reported represent the observed range across these repeated experiments rather than individual measurements.

TABLE III
MEASURED SYSTEM LATENCIES

Operation	Latency
Speech-to-Text	1–3 seconds
Translation	0.5–2 seconds
Text-to-Speech	1–2 seconds
AR Initialization	2–4 seconds
Route Computation	< 1 second
Total Translation Pipeline	4–7 seconds

Latency performance was good when the network was stable. Network condition variations had the most influence on overall performance, as several cloud-based operations run in sequence. This affects user trust, particularly under time pressure, and is a known tradeoff of the cloud-dependent architecture.

VI. DISCUSSION AND CONCLUSION

CareNav demonstrates viability as a cohesive system integrating speech translation with AR indoor navigation, offering better usability than individual tools. Real-world testing at PES Hospital confirmed that all components worked together effectively: speech was captured, translated between English and Kannada, synthesized, and paired with ARCore-based navigation instructions in an end-to-end pipeline.

The most notable constraints stem from the environment and operational conditions rather than algorithmic limitations. The AR tracking system was stable in most sessions but drifted in surroundings featuring repetitive visual patterns and reflective surfaces [5]. This is acknowledged as a known limitation of vision-based localization and is identified as a priority area for future improvement. Internal testing showed that users gained greater confidence in their navigational ability, and hospital staff reported fewer requests for directional assistance. Integrated support reduced friction noticeably in outpatient departments, diagnostic wings, and reception areas.

The system's current limitations include its reliance on stable network connectivity — a design assumption appropriate for modern hospital facilities and consistent with the infrastructure at the deployment site. Offline resilience remains a future enhancement goal. General translation engines continue to struggle with highly specialized medical narratives; improving domain adaptation through medical corpus fine-tuning is the most impactful next step. AR tracking robustness in corridors with reflective surfaces and multi-floor transition reliability are also priority areas for improvement before broader clinical deployment.

VII. FUTURE WORK

Future efforts will address key limitations before wider clinical implementation. On-device neural machine translation would remove network dependency and reduce the existing 4–7 second translation latency. Fine-tuning on medical corpora and a curated Kannada medical terminology database would improve translation accuracy for clinical communication. Multi-modal sensor fusion incorporating Wi-Fi fingerprinting, Bluetooth beacons, inertial measurements, and barometric pressure would stabilize AR tracking in areas with poor visual conditions, including reflective corridors. Robustness improvements for AR localization in ambiguous visual environments are a direct response to limitations observed during real-world testing. Extending language support to Hindi, Tamil, and Telugu would cover approximately 70 percent of South and Central Indian citizens. Randomized controlled trials across multiple hospitals with a minimum of 500 patients would provide statistically validated evidence of clinical effectiveness and surface deployment-specific requirements.

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